



Profiling the Activity of the Potent and Highly Selective CDK2 Inhibitor BLU-222 Reveals Determinants of Response in *CCNE1*-Aberrant Ovarian and Endometrial Tumors

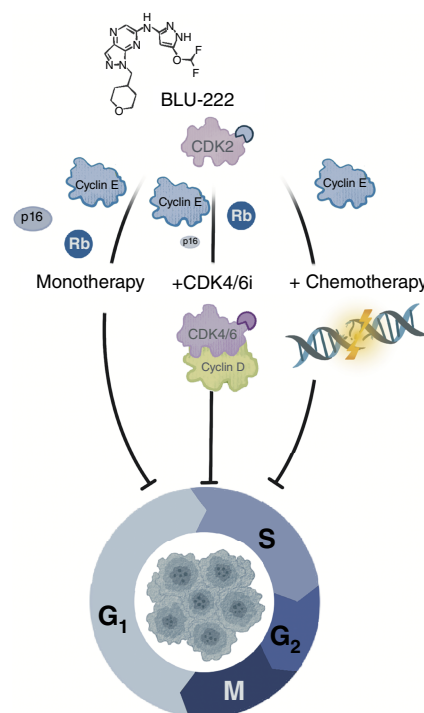
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ABSTRACT

BLU-222 is an investigational, potent, highly selective, orally bioavailable cyclin-dependent kinase 2 (CDK2) inhibitor in clinical development. BLU-222 demonstrated robust antitumor activity in select *CCNE1*-high ovarian and endometrial cancer models. We used a combination of CRISPR whole-genome screens coupled with targeted genetic and pharmacologic approaches in ovarian and endometrial cell lines to identify biological determinants to predict BLU-222 monotherapy activity. Rb and p16 expression were biomarkers that enriched for CDK2-dependency/BLU-222 sensitivity in *CCNE1*-overexpressed, non-amplified cells. Furthermore, intact Rb and low p16 expression predicted a BLU-222 and CDK4/6 inhibitor combination response. BLU-222 demonstrated robust activity in combination with carboplatin or paclitaxel in *CCNE1*-aberrant models, rendering chemotherapy-resistant tumors strongly sensitive to the combination. These findings demonstrate that response to CDK2 inhibition by BLU-222 can be further predicted using a combinatorial biomarker signature that could refine patient selection criteria in *CCNE1*-high patients and support clinical development.

Significance: The identification of biomarkers of response to the CDK2-selective inhibitor BLU-222 and effective combinations with CDK4/6 inhibitors or chemotherapy could enable precision medicine strategies for CDK2 inhibition in ovarian and endometrial cancer.

See related article by Dommer and colleagues, p. 1310



Introduction

Amplification of *CCNE1*, encoding for cyclin E1, is a common oncogenic event in ovarian and endometrial cancers, reported in up to ~20% of ovarian cancers and ~40% of uterine carcinosarcomas (1). Aberrant, high expression of *CCNE1* has been associated with chemoresistance, poor prognosis, and lower disease-free survival

(1–3). Novel therapeutics are therefore needed to improve the treatment response rate and survival in these cancers.

Cyclin E1 is an activating subunit of cyclin-dependent kinase 2 (CDK2), which together regulate the G₁–S-phase transition through phosphorylation/inactivation of the tumor suppressor, retinoblastoma protein 1 (Rb; ref. 4). In normal mammalian cells, activation of CDK2 is regulated by precise modulation of cyclin E levels, first by CDK4/6–cyclin D–induced activation of E2F signaling, including the synthesis of cyclin E–encoding genes (*CCNE1* and *CCNE2*) in G₁. This is followed by ubiquitin–proteasome pathway–mediated degradation of cyclin E during the late S-phase (2). CDK2 activity is also negatively regulated by kinase inhibitory protein/CDK-interacting proteins (p21, p27, and p57; ref. 5). *CCNE1* amplification and cyclin E1 overexpression drive aberrant CDK2 activity and dysregulate the cell cycle (6). High cyclin E1 levels are associated with increased DNA double-stranded breaks, genomic instability, and dysregulation of genes involved in cellular survival and proliferation (1, 7). *CCNE1*-amplified cancer cell lines exhibit mutual exclusivity with breast cancer (BRCA) gene mutations and insensitivity to PARP inhibitors (8) but are dependent on CDK2 for

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proliferation (9, 10) and are sensitive to CDK2 inhibitors (CDK2i; refs. 7, 11, 12). CDK2 is an attractive target for CDK2-vulnerable cancers and warrants further clinical development.

BLU-222 is an investigational CDK2i with strong antitumor activity as monotherapy in preclinical *CCNE1*-amplified ovarian cancer models. Here, we identified additional biological determinants to predict BLU-222 monotherapy activity in *CCNE1*-high ovarian and endometrial cancers and demonstrated the use of expression of these biomarkers in predicting response when combined with CDK4/6 inhibitors (CDK4/6i; e.g., ribociclib). Furthermore, we demonstrated that BLU-222 can be combined with chemotherapeutic agents for a strong antitumor effect, acting as a chemosensitizer in the *CCNE1*-aberrant, chemotherapy-refractory setting.

Materials and Methods

Cell lines and culture

Cell line and culturing information is in Supplementary Table S1. Stable cell line generation is in the Supplementary Methods. Cell lines were routinely tested for *Mycoplasma* and identity was confirmed by short tandem repeat profiling (approximately every three passages).

Identifying and evaluating biomarker signatures in publicly available data

We retrieved data from the Cancer Cell Line Encyclopedia (CCLE) Cancer Dependency Map (DepMap; RRID: SCR_017655) 23Q4 public release and tested the correlation between the CRISPR gene effect (Chronos score) or RNAi gene effect (DEMETER2 score) with copy number (CN) and mutation data from whole-exome sequencing or gene expression data from mRNA sequencing. The data from DepMap 23Q4 were also used to evaluate the relationship between genomic and transcriptomic markers and treatment. The Kruskal–Wallis test was used for correlations between categorical and continuous variables whereas the Wilcoxon rank-sum test was used for correlations between binary and continuous variables.

Patient population biomarkers were evaluated using publicly available data from The Cancer Genome Atlas (RRID: SCR_003193).

Western blotting

Lysates were prepared using PhosphoSafe Lysis Buffer (EMD Millipore, 71296), and Western blots were performed using standard methods. Antibodies used were cyclin E1 1:500 (SC-247; RRID: AB_627357), p16 1:1,000 (CST 80772; RRID: AB_2799960), CDK2 1:1,000 (SC-6248; RRID: AB_627238), CDK4 1:1,000 (CST 12790; RRID: AB_2631166), cyclin D1 1:1,000 (SC-450; RRID: AB_627342), TK-1 1:1,000 (CST 28755; RRID: AB_3668643), Rb 1:1,000 (CST 9309; RRID: AB_823629), pRbS780 1:1,000 (CST 9307; RRID: AB_330015), pRbS807/811 1:1,000 (CST 8516; RRID: AB_11178658), pRbT821/826 1:500 (SC-271930; RRID: AB_10655683), actin (R) 1:2,000 (CST 8457; RRID: AB_10950489), and actin (M) 1:5,000 (CST 3700; RRID: AB_2242334). Secondary antibodies (LI-COR 926-33212, RRID: AB_621847; 926-33213, RRID: AB_621848; 926-68070, RRID: AB_10956588; 926-68071, RRID: AB_10956166; and 926-68076, RRID: AB_10956590) were used at a 1:10,000 dilution, and signals were quantified using LI-COR Odyssey CLx Image Studio (RRID: SCR_014579).

IncuCyte proliferation

Cells were seeded at a starting confluency of 5% to 10% and dosed with BLU-222 or transfected with 25 nmol/L ON-TARGETplus Non-targeting Control Pool siRNA or ON-TARGETplus Human

CDK2 siRNA SMARTPool (Horizon Discovery D-001810-10/L-003236-00) with Lipofectamine 2000 (Invitrogen 11668027). Percent confluency was measured until controls reached 100% confluency or plateaued, as calculated using IncuCyte software (Sartorius, RRID:SCR_023147).

CyQUANT proliferation assay

Cells were seeded in black, clear-bottom, 384-well (Corning CLS3571) or 96-well (Corning 3904) plates at densities listed in Supplementary Table S1. For single-agent treatment, the cells were treated in a 10-point dose titration, with a top dose of 25 μ mol/L at 1:4 dilutions. For combination treatment, the cells were treated in a 10-point dose titration matrix, with a top dose of 5 μ mol/L at 1:4 dilutions. CyQUANT Direct Proliferation Assay was performed according to the manufacturer's instructions (Invitrogen, C35012) after 120-hour treatment. Percent inhibition was determined relative to DMSO and staurosporine controls to determine IC₅₀ values. Percent inhibition relative to DMSO and day 0 was calculated to determine half-maximal growth inhibition (GI₅₀) values. Synergy scores were determined using SynergyFinder 3.0 (RRID: SCR_019318; ref. 13). Cell line proliferation screens were run at ChemPartner (RRID: SCR_026139).

Cell-cycle profiling

Cells at ~65% confluency were treated with BLU-222 for 22 hours and pulsed with 10 μ mol/L EdU for 2 hours. The cell-cycle phase was detected by flow cytometry using the Click-iT EdU Alexa Fluor 488 Flow Cytometry Kit or Alexa Fluor 647 kit (Invitrogen C10420/C10634) following the manufacturer's instructions. FACS analysis was performed using LSRFortessa (BD Biosciences) and analyzed using FlowJo v10.8.1 software (BD Life Sciences; RRID: SCR_008520).

Clonogenic assays

Cells were treated at indicated compound doses, and culture medium and compounds were refreshed every 4 days until day 15. The cells were then stained with 0.125% crystal violet (Sigma-Aldrich, V5265). The surviving fraction of treated cells was calculated relative to DMSO control using ImageJ (64-bit, Java 8; RRID: SCR_003070).

AlphaLISA assays

Rb phosphorylation was determined using the AlphaLISA SureFire Ultra Phospho-Rb (Thr821/826) Detection Kit and/or the AlphaLISA SureFire Ultra Human and Mouse Phospho-Rb (Ser807/811) Detection Kit (Revvity ALSU-PRB-B50K/ALSU-PRB-A50K) in cells dosed for 18 hours. Phospho-Lamin S22 was determined using the AlphaLISA SureFire Ultra Human Phospho-Lamin A/C (Ser22) Detection Kit (Revvity ALSU-PLAM-A50K) in cells dosed for 2 hours. OVCAR-3 cells were maintained in RPMI medium with 20% FBS and plated in serum-free DMEM at 25,000 cells per well in a 96-well plate (Corning 353296) for 30 hours. Media were replaced with DMEM with 10% FBS, and the cells were treated with compound in a 9-point dose titration, with a top dose 10 μ mol/L at 1:4 dilution. AlphaLISA was performed according to the manufacturer's instructions (Revvity). IC₅₀ was determined in GraphPad Prism (RRID: SCR_002798) using nonlinear regression, variable slope, four-parameter fit. For tumor analysis, 5 μ g of the tumor lysate prepared in PhosphoSafe (EMD Millipore, 71296) was used for AlphaLISA.

CRISPR/Cas9 resistance screen

OVCAR3-Cas9 cells (~100M) were transduced at a multiplicity of infection = 0.5 with 80K single-guide RNA (sgRNA)

CRISPR-knockout (KO) lentiviral library (Cellecra KOHGW-80K-P). Three days after infection, 2 µg/mL puromycin was added to cell media to select for transduced cells for 14 days. The cells were then treated for 7 days with DMSO or 200 nmol/L BLU-222, with ~50M cells per sample in duplicate. sgRNA enrichment was detected by next-generation sequencing. Raw sgRNA counts were scaled to total sgRNA for each gene per sample; fold change against plasmid was calculated, and Z-transform was applied. The redundant siRNA activity (RSA) method was used to estimate RSA scores for each gene per sample (14). The minimum value of either the fifth or 95th percentile of the normalized RSA score distribution was used as the significance threshold, and the difference in Z-score between the BLU-222-treated gene and the DMSO-treated gene quantified the direction and magnitude of effect.

Animal studies

In vivo cell-derived xenograft (CDX) studies were performed at Shanghai ChemPartner or Pharmaron, and patient-derived xenograft (PDX) studies were performed at XenoSTART according to the protocols approved by the Institutional Animal Care and Use Committee at each institution. For all mouse studies, animals were monitored throughout the study period for irregular behavior (e.g., changes in mobility, food and water consumption, body weight, eye/hair matting, or any other abnormal condition), and any mortality and/or abnormal clinical signs were recorded. Detailed study information is in the Supplementary Methods.

Fine chemicals

BLU-222 was synthesized at Blueprint Medicines. Ribociclib was sourced from MedChemExpress (HY-15777), carboplatin from Selleck Chemicals (S1215), paclitaxel from Selleck Chemicals (S1150), and gemcitabine from Selleck Chemicals (S1714).

Data availability

Publicly available RNA sequencing (RNA-seq) and whole-exome sequencing data from CCLE DepMap (<https://depmap.org/portal/>; RRID: SCR_017655) and The Cancer Genome Atlas (<https://www.cancer.gov/ccg/research/genome-sequencing/tcga>; RRID: SCR_003193) were used to support the findings of this study.

The RNA-seq data generated in this study are available in Gene Expression Omnibus at GSE282041, and a reproducible capsule is hosted on the Code Ocean platform (<https://codeocean.com/capsule/0036296/tree/v1>). All other raw data generated in this study are available upon request from the corresponding author.

Results

CCNE1 amplification predicts sensitivity to the CDK2 inhibitor BLU-222 in ovarian and endometrial cancer cells

Ovarian and endometrial cancers have a high prevalence of *CCNE1* amplification (Supplementary Fig. S1A) and have similar clinical, pathologic, and molecular features (15). In ovarian and endometrial cell lines, *CCNE1* CN amplification (CN ≥6) strongly correlated with CDK2 dependency in DepMap (Supplementary Fig. S1B–S1K; ref. 16), consistent with previous reports (17, 18).

BLU-222 is a potent, highly selective ATP-competitive CDK2i (Fig. 1A and B; ref. 19). Enzymatic and live-cell target engagement measured by NanoBRET demonstrated that BLU-222 achieved nanomolar potency on CDK2 with strong selectivity over CDK

family members (1/4/6/7/9; Fig. 1C; Supplementary Table S2). BLU-222 has excellent kinome selectivity with an S(10) of 0.045, measured by KINOMEScan at 3 µmol/L (Fig. 1B and C). Importantly, in the CDK2-dependent cell line OVCAR-3, BLU-222 displays nanomolar cellular potency on phosphorylated Rb (pRb) Thr821/826 with strong selectivity over pLamin S22, a measure of CDK1 activity (Fig. 1D). Furthermore, BLU-222 spared CDK4 in cells, as measured by pRb Ser807/811 inhibition in T47D, a CDK4-dependent cell line (Fig. 1E).

BLU-222 potently and dose-dependently inhibited proliferation in *CCNE1*-amplified ovarian/endometrial cancer cell lines (Fig. 2A) while sparing *CCNE1*-normal (diploid, mRNA not overexpressed) cell lines (Fig. 2B). We confirmed CDK2 sensitivity of *CCNE1*-amplified cell lines by knockdown of CDK2 (Supplementary Fig. S2A–S2D), consistent with published data (9, 17). To evaluate the *CCNE1* CN as a predictive biomarker for BLU-222 response, we tested a panel of ovarian and endometrial cancer cell lines ($n = 42$) that harbored *CCNE1* amplification (CN ≥6), *CCNE1* CN gain (CN = 3–5), or diploid *CCNE1* CN (<3; Supplementary Table S1). The strong responder GI₅₀ threshold of 200 nmol/L was determined through a Gaussian mixture model analysis fitted by the Expectation–Maximization algorithm, based on the measured GI₅₀s of all treated cell lines. At this dose in OVCAR-3, pRb Thr821/826 inhibition (CDK2 activity measure) plateaus and pLamin S22 is not significantly inhibited (Fig. 1D). *CCNE1* amplification was associated with a strong BLU-222 antiproliferative response, whereas *CCNE1* CN gain demonstrated variable BLU-222 sensitivity, and diploid *CCNE1* CN cell lines were largely BLU-222 insensitive (Fig. 2C).

We investigated whether high *CCNE1* mRNA could predict BLU-222 sensitivity. The threshold for *CCNE1* overexpression was set at the top quartile of mRNA expression from CCLE data. *CCNE1* overexpression alone did not enrich for BLU-222 sensitivity (Fig. 2D). Therefore, although *CCNE1* CN amplification strongly predicted sensitivity to BLU-222, high *CCNE1* mRNA expression alone was insufficient.

We next evaluated BLU-222 in preclinical ovarian cancer CDX models. OVCAR-3 is a *CCNE1*-amplified (CN >10), high-grade serous ovarian cancer (HGSOC) cell line. To develop the OVCAR-3 T2A model, OVCAR-3 tumors were passaged once through mice to improve *in vivo* growth kinetics. BLU-222 was well tolerated in mice and induced stasis in OVCAR-3 T2A at doses in which sustained inhibition of pRb Thr821/826 and pRb Ser807/811 was achieved for at least 12 hours (Fig. 2E; Supplementary Fig. S3A–S3D). ES-2, a *CCNE1*-normal (CN <3, mRNA normal), ovarian clear cell carcinoma CDX was insensitive to BLU-222 (Fig. 2F), and pRb was not inhibited (Supplementary Fig. S3E–S3H). This confirmed that CDK2 inhibition suppresses tumor growth in a *CCNE1*-amplified model *in vivo* and that antitumor activity is correlated with pRb inhibition, as reported with other CDK2i (12, 20–22).

BLU-222 disrupts Rb signaling in *CCNE1*-amplified cells to arrest cell-cycle progression

To promote the G₁–S transition, CDK4/6-cyclin D and CDK2-cyclin E phosphorylate Rb in late G₁, leading to Rb inactivation and release of the E2F transcription factors that promote S-phase-related gene expression, including TK1 (23, 24). Upon treatment with BLU-222, both pRb and TK1 were inhibited in *CCNE1*-amplified cells but not in *CCNE1*-normal cells (Fig. 2G and H). Consistent with the role of CDK2 promoting G₁–S transition in CDK2-dependent cell lines (12, 25), BLU-222 led to G₁ arrest in *CCNE1*-amplified cells (Fig. 2I; Supplementary Fig. S4A–S4D). In

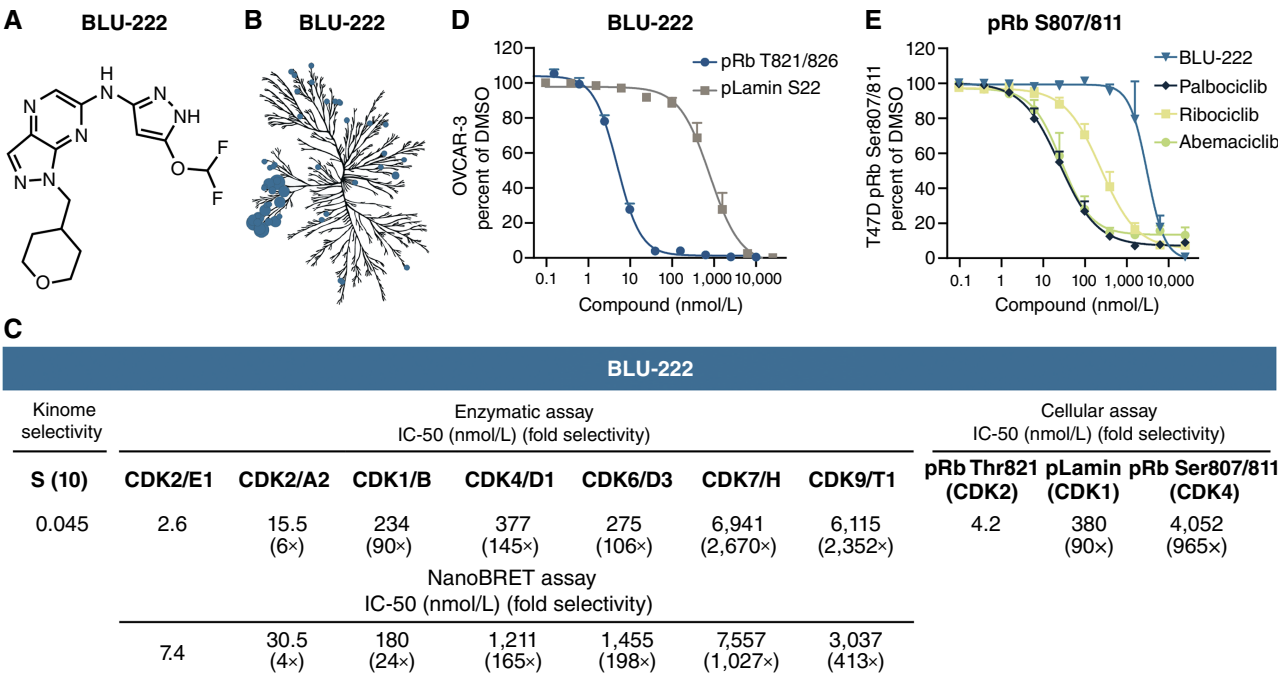


Figure 1. BLU-222 is a potent and selective CDK2i. **A**, Chemical structure of BLU-222. **B**, BLU-222 kinome tree. **C**, Enzymatic, NanoBRET, and cellular potency and selectivity of BLU-222. Kinome selectivity tested at 3 μ mol/L by KINOMEScan; S(10) is the number of kinases inhibited at <10 percent of control divided by the total number of human wild-type kinases. Biochemical assays were conducted at 1 mmol/L ATP in the presence of appropriate cyclin (indicated). Fold selectivity is noted in brackets. **D**, CDK2 cellular potency measured by inhibition of pRb Thr821/826 by AlphaLISA after 18-hour dosing in OVCAR-3. pLamin S22 inhibition was used as a cellular readout of CDK1 inhibition by AlphaLISA after 2-hour dosing. **E**, CDK4/6 cellular potency in T47D, a CDK4/6-dependent breast cell line. In T47D, pRb Ser807/811 is controlled by CDK4/6 and therefore was used as a readout for CDK4/6 activity after 18-hour dosing.

some cell lines, BLU-222 at higher doses induced accumulation in G₂/M. Similar to other CDK inhibitors, this may be due to inhibition of the CDK2-cyclin A complex or of CDK1, which occurs at concentrations of BLU-222 well above that needed to selectively inhibit CDK2-cyclin E (Fig. 1C and D). BLU-222 did not significantly affect cell-cycle distribution in *CCNE1*-normal cells (Fig. 2J; Supplementary Fig. S4E–S4H). We also evaluated the cell-cycle distribution in BLU-222-treated *CCNE1*-overexpressed, CN <6 cell lines. Interestingly, in BLU-222-responder cell lines (GI₅₀ <200 nmol/L), BLU-222 resulted in a G₁ or G₂-M arrest while the cell-cycle distribution was not affected in *CCNE1*-overexpressing, BLU-222-nonresponder cell lines (Supplementary Fig. S5). Together, pRb inhibition, TK1 downregulation, and accumulation in G₁ suggest that BLU-222 inhibits the Rb–E2F axis, leading to an antiproliferative response.

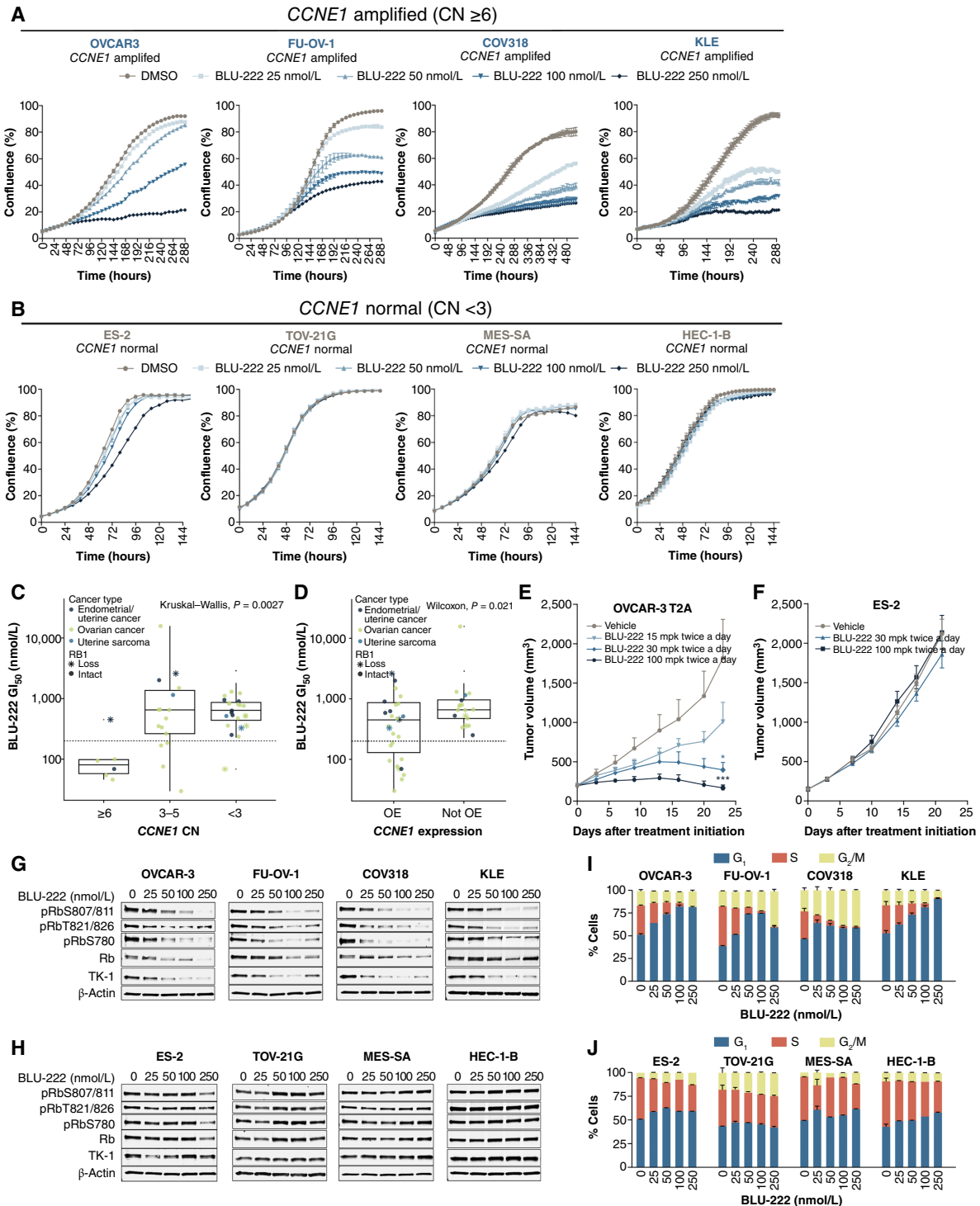
***RB1* (Rb) and *CDKN2A* (p16) are essential for BLU-222 sensitivity in *CCNE1*-amplified OVCAR-3**

To identify genes that modulate the BLU-222 antiproliferative response in *CCNE1*-amplified HGSOC, we performed a genome-wide CRISPR/Cas9 pooled knockout screen in OVCAR-3 cells (Supplementary Fig. S6A and S6B). OVCAR-3 was sensitive to BLU-222 both *in vitro* and *in vivo* (Fig. 2A and E), and therefore, gene knockouts that confer a change in BLU-222 activity represent markers for response. Five genes were identified that, when deleted, led to significant loss of BLU-222 sensitivity in OVCAR-3 cells: *RB1*, *CDKN2A*, *SAFB*, *KLH14*, and *HLA-DQA1* (Fig. 3A). We validated loss of BLU-222 sensitivity in the top hits: *RB1*, which encodes the

retinoblastoma protein, and *CDKN2A*, specific to the p16 transcript variant, the INK4 family member and endogenous inhibitor of CDK4/6 (26). In OVCAR-3, CRISPR/Cas9, knockouts were generated for *RB1* or *CDKN2A*; the two sgRNAs that achieved the greatest target loss were selected for further evaluation (Supplementary Fig. S6C and S6D). In OVCAR-3, knockout of *RB1* or *CDKN2A* led to loss of BLU-222 sensitivity and G₁ arrest (Fig. 3B and C; Supplementary Fig. S6E–S6H). Loss of BLU-222 sensitivity upon *RB1* or *CDKN2A* deletion was also confirmed in the *CCNE1*-amplified endometrial cell line, KLE (Supplementary Fig. S6I and S6J). Interestingly, nearly all *CCNE1*-amplified cell lines in CCLE were positive for Rb and had high p16 expression (Supplementary Table S3). These markers converged on modulation of the Rb pathway as critical for CDK2 sensitivity and BLU-222 response in HGSOC.

Rb and p16 are biomarkers of response to BLU-222 in *CCNE1*-high cells

Cells with high *CCNE1* mRNA expression without gene amplification (CN <6) had variable response to BLU-222 (Fig. 2D). Because Rb or p16 loss were resistance markers in the *CCNE1*-amplified setting, we explored whether endogenous Rb and/or p16 expression could be used as biomarkers of response in *CCNE1*-high ovarian and endometrial cell lines, regardless of *CCNE1* CN (Supplementary Fig. S7A–S7D). Cell lines were categorized using endogenous Rb and p16 expression detected by Western blotting, as RNA and protein expression by Western blotting were not strongly correlated. Rb intact was defined as 10th percentile or above of protein expression and no loss-of-function mutations. p16 high was defined as

**Figure 2.**

CCNE1-amplified ovarian and endometrial cancer cells are sensitive to BLU-222. **A** and **B**, Cell confluency measured by IncuCyte in BLU-222-treated *CCNE1*-amplified (CN ≥ 6 , mRNA high; **A**) or *CCNE1*-normal (CN <3, mRNA normal; **B**) cell lines. **C** and **D**, BLU-222 antiproliferative effect measured by CyQUANT after 5-day treatment. Each dot represents the BLU-222 GI_{50} of a single cell line. **C** and **D**, Cells were categorized by *CCNE1* CN (**C**) or *CCNE1* (**D**) mRNA expression derived from CCLE. The BLU-222 strong responder threshold was $GI_{50} \leq 200$ nmol/L. OE, overexpressed. **E** and **F**, *In vivo* growth kinetics of xenograft models in response to BLU-222. Mean tumor volume $\pm$ SEM is plotted for OVCA3 T2A (*CCNE1*-amplified, Rb-intact, p16-high expressor; **E**) in NOD/SCID female mice and ES-2 (*CCNE1* normal; **F**) xenograft in BALB/c nude female mice. Statistical deviation from vehicle assessed by two-way ANOVA. *, $P < 0.05$; ***, $P < 0.001$. **G** and **H**, Western blots for indicated proteins in BLU-222-treated cells (24 hours): *CCNE1*-amplified (**G**) and *CCNE1*-normal (**H**) cell lines. **I** and **J**, Cell-cycle phase analysis after 24-hour BLU-222 treatment in *CCNE1*-amplified (**I**) and *CCNE1*-normal (**J**) cell lines.

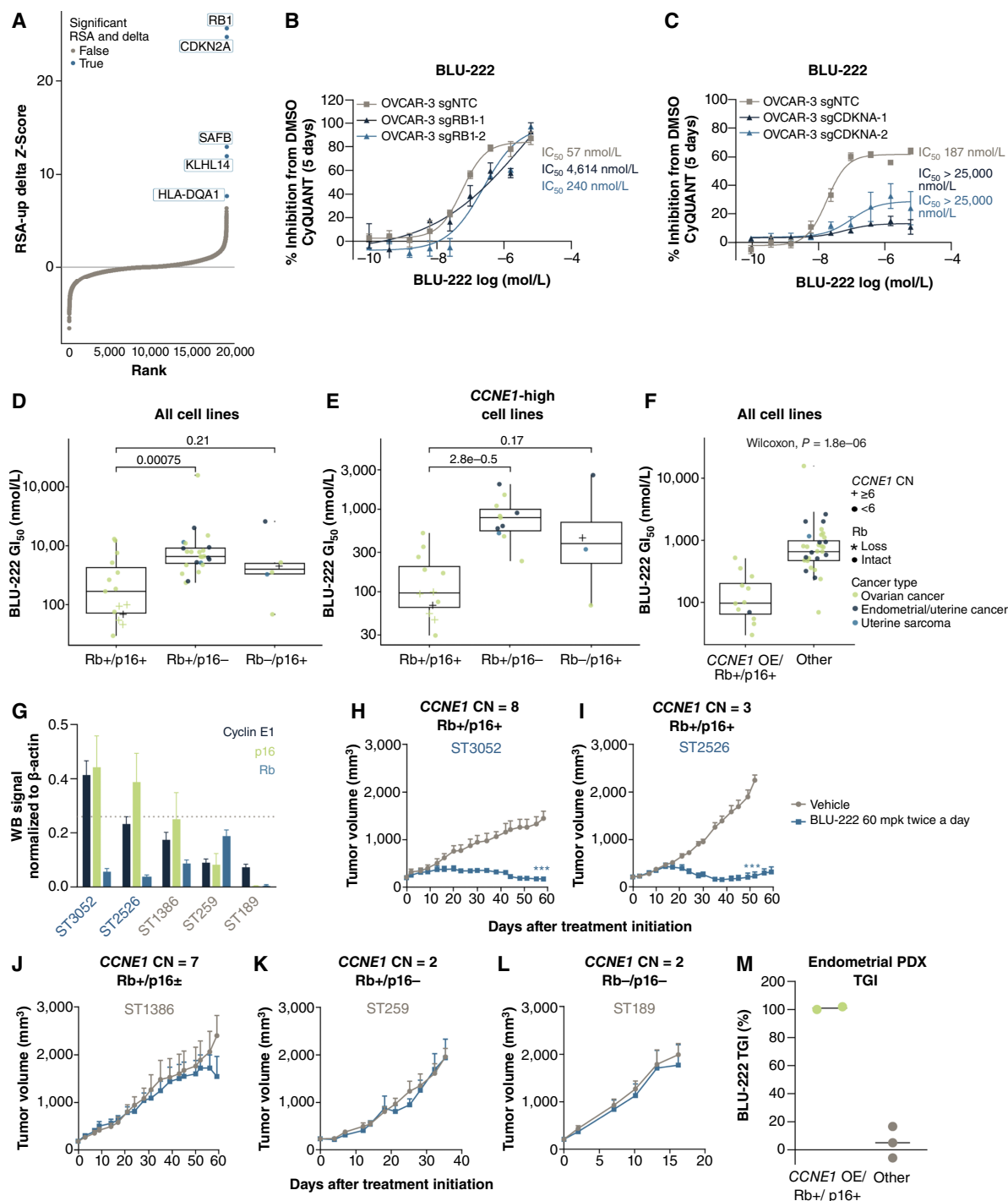


Figure 3.

Rb and p16 are determinants of response to BLU-222 in *CCNE1*-high cells. **A**, A CRISPR whole-genome library deletion screen was used to identify markers of resistance to BLU-222 in OVCAR-3 cells by monitoring outgrowth in the presence of BLU-222 (200 nmol/L). Resistance screen results represented by a waterfall plot of gene dependency Z-scores. Markers with strong magnitude of effect that reached statistical significance (t test $P < 0.05$) are marked in blue. **B** and **C**, BLU-222 antiproliferative response was assessed by 5-day CyQUANT in OVCAR-3 sgRB1 lines (**B**) and sgCDKN2A lines (**C**). **D–F**, BLU-222 GI_{50} s from cell lines tested in **C** were categorized by *CCNE1* mRNA and Rb/p16 protein expression status as measured by Western blotting. **G**, Western blot (WB) quantifications of cyclin E1, Rb, and p16 in endometrial PDX study vehicle tumors. Representative Western blot image in Supplementary Fig. S10. Blue font, predicted responders to single-agent BLU-222; gray font, predicted nonresponders. **H–L**, *In vivo* growth kinetics in response to BLU-222 60 mpk twice daily in *CCNE1*-high uterine PDX models. Rb and p16 protein expression status as indicated. Mean tumor volume $\pm$ SEM is plotted. **M**, TGI in endometrial PDX models categorized by cyclin E1, Rb, and p16 status based on protein expression quantification in **G**.

above median protein expression (Supplementary Fig. S7A and S7B). Loss of Rb or p16 trended toward nonresponse to BLU-222 (Fig. 3D). When focused on *CCNE1*-high cell lines, Rb-intact and high p16 cells were BLU-222 responders, and the cells with p16 loss or Rb loss were nonresponders to BLU-222 (Fig. 3E). Selecting cells using all three biomarkers, the *CCNE1*-high/Rb-intact/p16-high signature strongly enriched for BLU-222 responders ($P = 1.8 \times 10^{-6}$, Wilcoxon rank-sum test; Fig. 3F), consistent with CDK2 dependency observed in DepMap/CCLE (Supplementary Fig. S7E–S7G).

RNA-seq analysis upon BLU-222 treatment in a panel of 18 BLU-222 responder and nonresponder ovarian cell lines demonstrated clustering of *CCNE1*-high/Rb-intact/p16-high cell lines. As expected, BLU-222 induced a reduction in E2F pathway genes in responder cell lines, as well as downregulation of G₂–M checkpoint and mitotic spindle pathways (Supplementary Fig. S8A–S8F). Ploidy, ubiquitination pathways, and FBXW7 expression or mutations were not correlated with BLU-222 response (Supplementary Figs. S8 and S9A–S9D). However, mutations in the ubiquitination pathway genes were associated with decreased BLU-222 sensitivity (Supplementary Fig. S9E). Analysis of genes co-amplified with *CCNE1* in BLU-222 responder cell lines revealed that these genes resided within the 19q12 amplicon, and therefore were unlikely independently contributing to CDK2 sensitivity (Supplementary Table S4). Combining the CRISPR screen data with the BLU-222 cell line proliferation panel, we generated a combinatorial biomarker signature to predict CDK2 dependency and BLU-222 sensitivity in ovarian and endometrial cancer cells with high *CCNE1* expression, even in the absence of *CCNE1* CN amplification.

To evaluate the contribution of these biomarkers in tumors, we tested BLU-222 monotherapy in endometrial PDX models (Supplementary Table S5). All models had high *CCNE1* mRNA levels, as defined by the top quartile of model baseline expression in the XenoSTART model database; ST2526, ST3052, and ST1386 are *CCNE1* CN aberrant. The models had varying levels of expression of Rb and/or p16 (Fig. 3G; Supplementary Fig. S10A). BLU-222 dosed at 60 milligrams per kilogram (mpk) twice daily resulted in pharmacokinetic exposure that was sufficient to predict single-agent activity in responder tumors, exceeding the OVCAR-3 cellular pRb IC₉₀ for at least 12 hours, and was well tolerated in mice (Supplementary Fig. S10B–S10M). ST3052 (*CCNE1* CN = 8, Rb⁺/p16⁺) and ST2526 (*CCNE1* CN = 3, Rb⁺/p16⁺) were strongly responsive to BLU-222, reaching stasis, and this activity was not significantly enhanced at 100 mpk BLU-222 (Fig. 3H and I; Supplementary Fig. S10N and S10O). ST1386 is *CCNE1*-amplified (CN = 7) but had lower levels of p16 expression that, relative to β-actin, were on the threshold for “p16 high” as determined by the *in vitro* cell line panel. This model was not responsive to BLU-222 at 60 mpk (Fig. 3J) but was responsive to 100 mpk (Supplementary Fig. S10P). ST259 and ST189 are *CCNE1* CN normal but high *CCNE1* mRNA expressors. ST259 has low levels of p16 expression and did not respond to BLU-222 monotherapy (Fig. 3K). ST189 lacked both Rb and p16 expression and did not respond to BLU-222 monotherapy (Fig. 3L). Therefore, in these *CCNE1*-high endometrial PDX models, *CCNE1* status combined with p16 and Rb status better predicted response to BLU-222 monotherapy than aberrations in *CCNE1* alone (Fig. 3M).

Inactivation of CDK4/6 renders p16-low cells vulnerable to CDK2 inhibition by BLU-222

CDKN2A (p16) alterations are common oncogenic events, and high p16 expression is a biomarker for CDK4/6i resistance (27–29).

We further investigated the role of p16 in modulating BLU-222 response using the OVCAR-3 isogenic *CDKN2A* CRISPR/Cas9 cells. *CDKN2A* deletion led to increased total and pRb and increased TK1, indicating functional upregulation of CDK4 activity compared with the control cells (Fig. 4A; Supplementary Fig. S11A and S11B). The sg*CDKN2A* knockout also led to increased CDK4-cyclin D1 association (Fig. 4B) and binding of CDK4/6i as demonstrated by cellular thermal shift assay (Supplementary Fig. S11C), consistent with restoration of CDK4 to an active state upon p16 loss in OVCAR-3. Reciprocally, induced overexpression of *CDKN2A* in OV-90 and AN3CA, p16-low, BLU-222–nonresponder cell lines sensitized to BLU-222 single agent (Fig. 4C–E) and resulted in G₁ arrest (Supplementary Fig. S11D and S11E). As expected, p16 overexpression in KLE, a p16-high, BLU-222–responder cell line, had no impact on response (Fig. 4C and F; Supplementary Fig. S11F).

We next tested whether pharmacologic inhibition of CDK4/6 could restore sensitivity to BLU-222 in the *CDKN2A*-KO cells. Using a clonogenic assay approach, OVCAR-3 sgNTC (nontargeting control sgRNA) and sg*CDKN2A* cell lines were treated with BLU-222 alone or in combination with ribociclib. Although BLU-222 + ribociclib conferred no additional antiproliferative effect compared with BLU-222 alone in the sgNTC line, BLU-222 sensitivity was restored in the sg*CDKN2A* cell lines in the presence of 1.5 or 3 μmol/L ribociclib (Supplementary Fig. S12A–S12E). BLU-222 treatment led to G₁ accumulation in control cells, with no additional effect upon cotreatment with ribociclib (Fig. 4G). In the *CDKN2A*-KO cells, BLU-222-induced G₁ accumulation was lost, and cotreatment of BLU-222 and ribociclib restored BLU-222-induced G₁ accumulation (Fig. 4G). These data are consistent with derepressed CDK4/6 compensating for the loss of CDK2 activity in regulating the G₁–S-phase transition. Depletion of CDK4, CDK6, or both in the sg*CDKN2A* cell lines indicate that both CDK4 and CDK6 activities must be suppressed to restore BLU-222 sensitivity in this setting (Supplementary Fig. S12F–S12J). Therefore, either CDK4 or CDK6 activity may drive resistance to CDK2i in ovarian cancer cells. Together, these data suggest a plasticity in CDK2 and CDK4/6 driving cell cycle progression, consistent with a report demonstrating CDK4/6 compensation upon CDK2i (30). We conclude that in a *CCNE1*-high setting in which CDK2 and CDK4/6 are active, both CDK2 and CDK4/6 must be inhibited to elicit an antiproliferative response.

We next asked whether p16 expression could be used as a biomarker for BLU-222 + ribociclib combination treatment broadly across *CCNE1*-high ovarian and endometrial cells. We posited that cell lines with high *CCNE1* expression and low p16 expression, and therefore intact CDK4/6 activity, would be sensitive to CDK2i in combination with CDK4/6i. Using synergy analysis on ovarian ($n = 34$) and uterine cell lines ($n = 13$), *CCNE1* expression alone was insufficient to predict BLU-222 + ribociclib combination response (Fig. 4H; Supplementary Table S1; Supplementary Fig. S13). Synergistic potential trended toward cell lines with intact Rb and low p16 expression (Fig. 4I), whereas the cells with high *CCNE1*, intact Rb, and low p16 expression were most predictive of BLU-222 + ribociclib synergistic effect (Fig. 4J).

To evaluate the contribution of p16 levels to BLU-222 + ribociclib effects *in vivo*, we used the OVK18 ovarian CDX model that is *CCNE1*-high/Rb-intact/p16-low. OVK18 is insensitive to BLU-222 and ribociclib single agents *in vitro*, but the agents demonstrated a strong synergistic potential (Fig. 5A and B). The BLU-222 + ribociclib combination had a robust antitumor effect in OVK18, greater

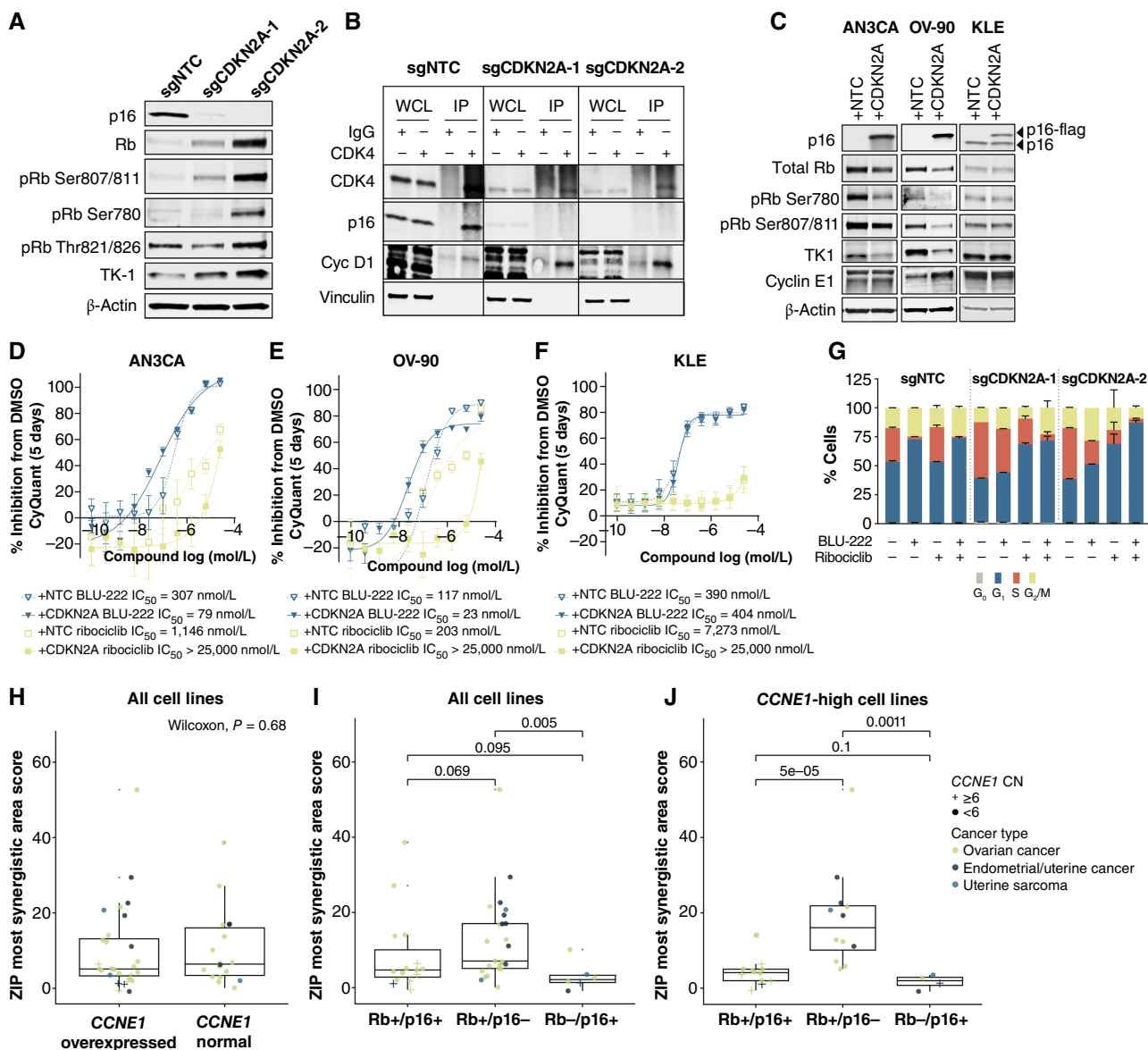


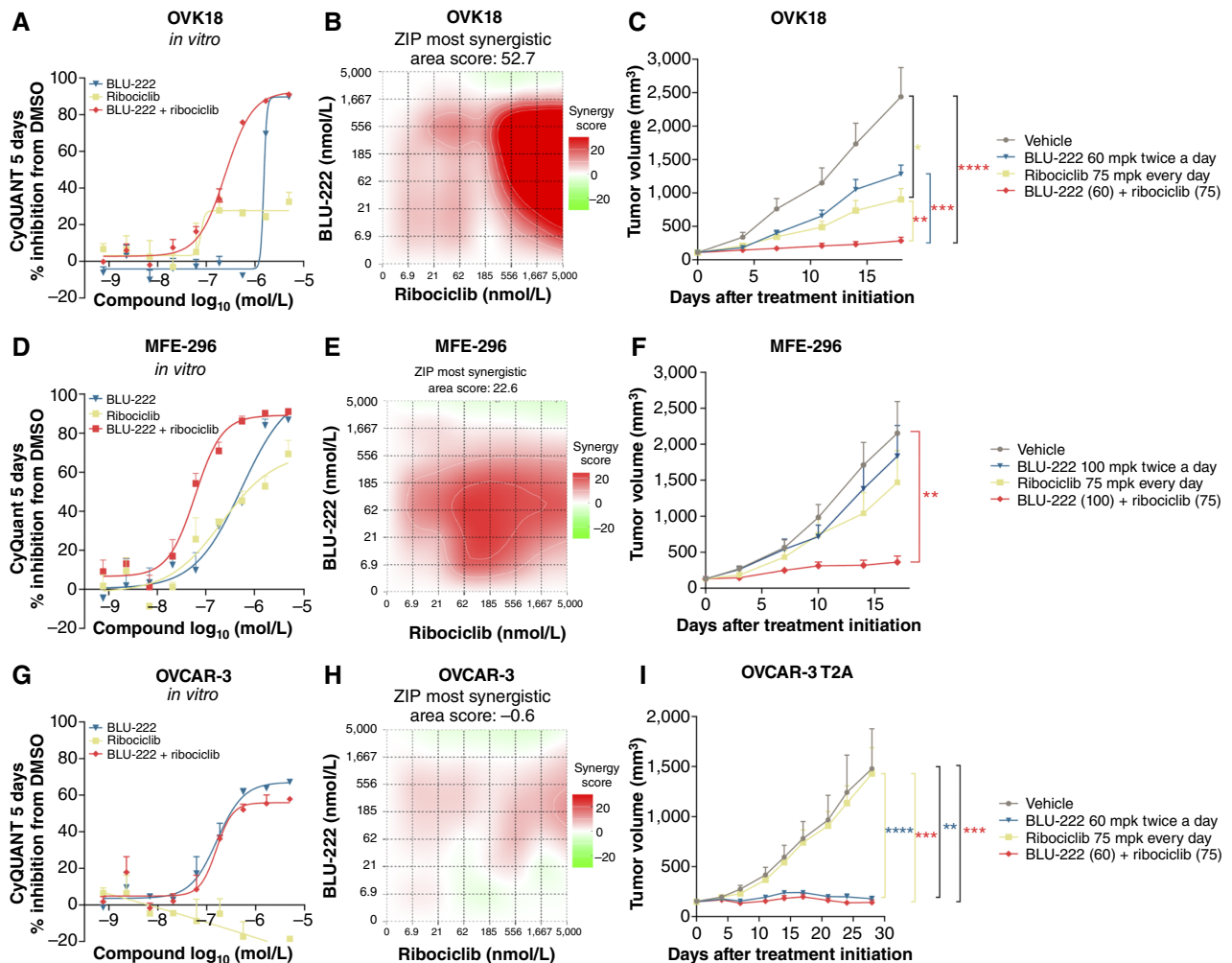
Figure 4.

Inhibition of CDK4/6 activity sensitizes *CCNE1*-high cells to BLU-222. **A**, Western blots for indicated proteins in OVCAR-3 sgCDKN2A deletion cell lines. Total Rb and pRb Ser807/811 were probed sequentially from one membrane; pRb Thr821/826 and pRb Ser 780 were probed sequentially from a second membrane. **B**, CDK4 coimmunoprecipitation demonstrates increased cyclin D1 association upon *CDKN2A* deletion in OVCAR-3. IP, immunoprecipitate; WCL, whole cell lysate. **C**, Western blots for indicated proteins in vector control (+NTC) or CDKN2A-overexpressing (+CDKN2A) cell lines. AN3CA and OV-90 express endogenously low levels of p16. KLE expresses endogenously high levels of p16 and is CDK2i sensitive. Total Rb and pRb Ser807/811 were probed sequentially from a single membrane. **D–F**, The antiproliferative effect of BLU-222 and ribociclib measured by CyQuant in +NTC or +CDKN2A cell lines. IC_{50} s are indicated. **G**, Cell-cycle analysis in *CDKN2A* isogenic OVCAR-3 cells treated with BLU-222 (250 nmol/L), ribociclib (3 μ mol/L), or both for 24 hours. **H–J**, Drug synergy analysis of BLU-222 and ribociclib in ovarian and uterine cell lines. *CCNE1*, Rb, and p16 designations determined as in Fig. 3.

than either single agent, and led to tumor stasis (Fig. 5C). Similar results were seen in MFE-296, a *CCNE1*-high/Rb-intact/p16-low endometrial cancer model (Fig. 5D–F). We then evaluated BLU-222 + ribociclib in OVCAR-3, a *CCNE1*-high/Rb-intact/p16-high model. In this setting, BLU-222 + ribociclib combination was not predicted to have additional benefit over BLU-222 alone because of pre-existing CDK4 inhibition via high levels of endogenous p16. As predicted by the *in vitro* data (Fig. 5G and H), the OVCAR-3 T2A

CDX model was refractory to ribociclib monotherapy, and BLU-222 + ribociclib combination showed no additional antitumor activity compared with BLU-222 alone (Fig. 5I). Furthermore, BLU-222 + ribociclib was well tolerated in mice, and the plasma exposure of each agent was unaffected in the combination (Supplementary Fig. S14).

Together, these data demonstrate that the combination of BLU-222 and a CDK4/6i has a strong antitumor effect in *CCNE1*-

**Figure 5.**

BLU-222 in combination with ribociclib induces strong antitumor activity in p16-low models. The BLU-222 + ribociclib combination effect was measured *in vitro* by 5-day CyQUANT and subsequent zero interaction potency (ZIP) synergy analysis and using *in vivo* CDX model tumor growth kinetics in OVK18 (A–C), a *CCNE1*-high, Rb-intact, and p16-low ovarian cell line; MFE-296 (D–F), a *CCNE1*-high, Rb-intact, and p16-low endometrial cell line; and OVCAR-3 (G–I), a *CCNE1*-high (*CCNE1*-amplified), Rb-intact, and p16-high ovarian cell line. Mean tumor volume is plotted $\pm$ SEM. Statistical deviation between groups as determined by two-way ANOVA at study endpoint. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$.

high, Rb-intact models when p16 levels are low. With the biomarker analysis for response to single-agent BLU-222 in cell lines, this suggests that in the low p16 setting, pharmacologically inactivating CDK4/6 induced a cellular state similar to settings in which p16 is expressed at high levels.

BLU-222 is a chemosensitizing agent in cyclin E1-high tumors

Given the encouraging combinability of BLU-222 and ribociclib that achieved strong antitumor effects in preclinical ovarian and endometrial models, we next assessed BLU-222 in combination with standard-of-care chemotherapeutic agents. *CCNE1* amplification is associated with homologous recombination proficiency and resistance to standard chemotherapeutic agents, such as platinum and taxane agents (1, 31). We previously reported that in *CCNE1*-amplified cells, CDK2 is critical for aberrant homologous recombination to stabilize collapsed replication forks, leading to DNA-damaging agent resistance (7). We therefore tested whether

inhibiting CDK2 in *CCNE1*-high tumors could restore sensitivity to chemotherapeutic agents.

To evaluate preclinical combination activity *in vivo*, BLU-222 was dosed with carboplatin or paclitaxel in the OVCAR-3 T2A CDX model. Both combinations were well tolerated (Supplementary Fig. S15A and S15B). As single agents, BLU-222, paclitaxel, or carboplatin demonstrated antitumor activity during the dosing period, but tumors grew after treatment cessation (Fig. 6A). BLU-222 in combination with carboplatin or paclitaxel induced robust tumor regression during the dosing period, which continued upon treatment cessation (Fig. 6A). To test BLU-222 combined with carboplatin or paclitaxel in a *CCNE1*-normal setting, we used the ES-2 CDX model. This model was refractory to BLU-222, carboplatin, and paclitaxel. BLU-222 + carboplatin or BLU-222 + paclitaxel had no additional antitumor effect (Fig. 6B).

We also measured BLU-222 + carboplatin activity in *CCNE1*-high ovarian PDX models. A *CCNE1* mRNA-high (CN = 2) PDX

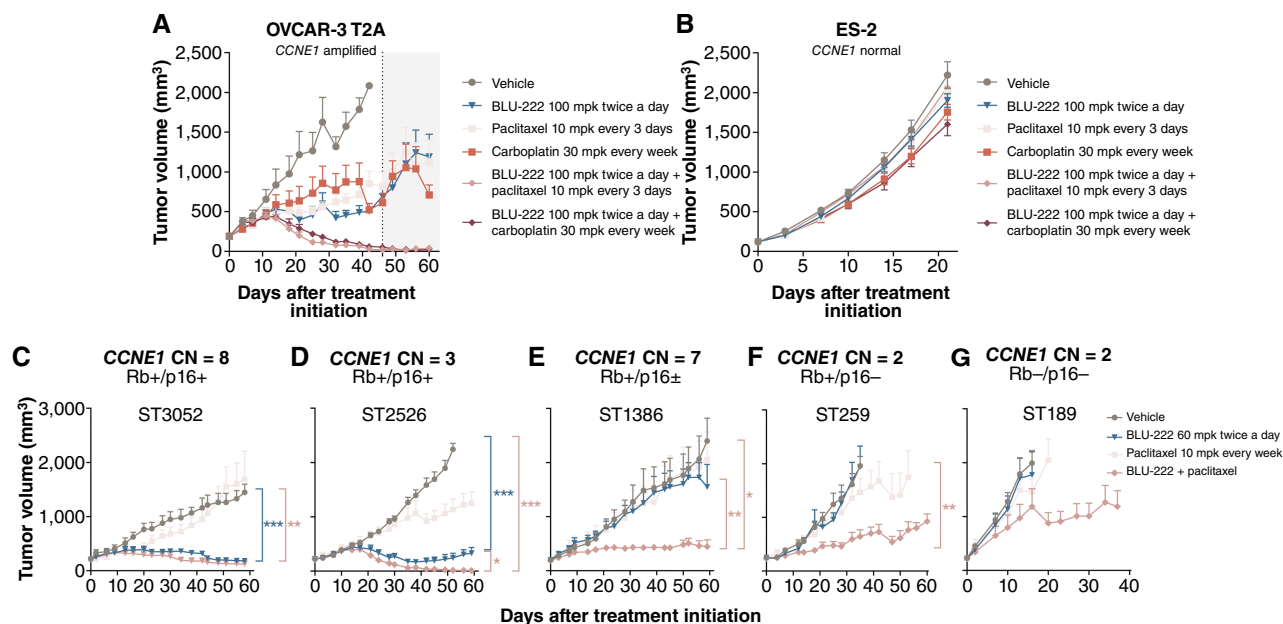


Figure 6.

BLU-222 in combination with chemotherapeutic agents demonstrates therapeutic benefit in ovarian and endometrial tumor models. **A** and **B**, *In vivo* growth kinetics in *CCNE1*-amplified OVCAR-3 T2A xenograft in NOD/SCID female mice dosed for 46 days (**A**) and ES-2 *CCNE1*-normal xenograft in BALB/c nude female mice dosed for 21 days (**B**). **C–G**, *In vivo* tumor growth kinetics in endometrial PDX models. Mean tumor volume $\pm$ SEM is plotted. Statistical deviation as determined by two-way ANOVA on latest day all the groups remained intact (ST3052, day 58; ST2526, day 52; ST1386, day 59; ST259, day 28; and ST189, day 13). *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

model derived from a treatment-naïve patient (ST270) was carboplatin sensitive [tumor growth inhibition (TGI) = 75.5%], with no additional antitumor benefit with BLU-222 + carboplatin (TGI = 65.2%; Supplementary Fig. S15C–S15J). In a *CCNE1*-amplified (CN = 8) PDX model derived from a carboplatin + paclitaxel-progressed patient (ST182B), combining BLU-222 with carboplatin significantly improved the single-agent antitumor effect, leading to tumor stasis (TGI = 71.4%; Supplementary Fig. S15C–S15J). Together, these data support BLU-222 in combination with chemotherapy in chemotherapy-resistant, CDK2-dysregulated tumors, such as in the *CCNE1*-amplified setting.

CCNE1 amplification and high levels of cyclin E1 are associated with increased levels of replication stress (32). In the *CCNE1*-amplified setting, preclinical treatment regimens that exacerbate replication stress and the DNA damage load, such as with ATR, Wee1, or PKMYT1 inhibition, can elicit strong antitumor effects (33, 34). We therefore tested BLU-222 with gemcitabine in the OVCAR-3 T2A model. Gemcitabine alone did not elicit a strong antitumor effect, but BLU-222 + gemcitabine induced a stronger response than either single agent. However, tumors regrew after treatment cessation (Supplementary Fig. S15K). We conclude that although BLU-222 + gemcitabine induced strong antitumor activity while on treatment, this combination is not as durable as BLU-222 combined with carboplatin or paclitaxel.

To evaluate biomarkers of response to BLU-222 + chemotherapy, we took a high-throughput *in vitro* approach in cell lines to measure potential synergy (Supplementary Fig. S16A–S16D; ref. 35). Interestingly, *CCNE1*, Rb, and p16 did not significantly predict synergistic potential (Supplementary Fig. S16B), suggesting that the chemotherapy combination response may be more broadly acting than the *CCNE1*-aberrant setting.

BLU-222 + paclitaxel induces strong antitumor response in *CCNE1*-high, Rb-intact endometrial PDX models

We evaluated the BLU-222 + chemotherapy combination in *CCNE1*-high endometrial PDX models (as tested in Fig. 3H–L). ST2526 was derived from a patient who had progressed on fluorouracil and radiation, whereas all remaining models were derived from treatment-naïve patients (Supplementary Table S5). All models were resistant to paclitaxel single-agent treatment (Fig. 6C–G). BLU-222 + paclitaxel was well tolerated, and the plasma exposure in combination was not affected from monotherapy treatment (Supplementary Fig. S10B–S10F and S10I–S10M). In the *CCNE1* CN-aberrant, Rb-intact, p16-high models (ST3052 and ST2526), BLU-222 induced tumor stasis (TGI = 102% and 107%, respectively; Fig. 6C and D). Although the combination with paclitaxel did not significantly improve from the antitumor activity of BLU-222 monotherapy in ST3052 (TGI = 107%; Fig. 6C), the combination led to tumor regression in ST2526 (TGI = 113%; Fig. 6D). A third *CCNE1* CN-aberrant, Rb-intact, but p16-low model (ST1386) did not respond to BLU-222 60 mpk (TGI = 17%) or paclitaxel (TGI = 9%) monotherapy; however, stasis was achieved with BLU-222 + paclitaxel combination (TGI = 78%; Fig. 6E). Therefore, in *CCNE1* CN-aberrant PDX models, BLU-222 + paclitaxel induced a strong antitumor response.

We next asked whether BLU-222 with paclitaxel would demonstrate a combination benefit in *CCNE1* mRNA-high but CN-normal models. ST259 (*CCNE1* CN 2, Rb⁺, and p16⁺) and ST189 (*CCNE1* CN 2, Rb⁺, p16⁺) were refractory to BLU-222 monotherapy (Fig. 6F and G). In ST259, a p16-low model, BLU-222 + paclitaxel had a strong antitumor effect (TGI = 80%; Fig. 6F). In ST189, a model lacking Rb expression, BLU-222 + paclitaxel led to tumor stabilization after 2 weeks (TGI = 52%; Fig. 6G). Together, these results revealed a combination benefit of BLU-222 + paclitaxel in

CCNE1-high models to overcome chemotherapy resistance, even beyond the *CCNE1* CN-amplified setting.

Discussion

With selective CDK2 inhibitors entering the clinic, identification of specific biomarkers to predict CDK2-vulnerable cancers is critical to enable selection of patients with target-responsive tumors. Ovarian, endometrial/uterine, gastric, and some breast cancers harbor high frequencies of *CCNE1* amplification and have been the focus of single-agent activity in preclinical and clinical investigation of CDK2i (Supplementary Table S2; refs. 1, 36). Here, we demonstrated robust preclinical activity of BLU-222 monotherapy in *CCNE1*-amplified ovarian and endometrial models (Figs. 2E, 3H, and I). Our study reveals aberrations in cancer cells that lead to a CDK2-dependent, dysregulated cell cycle. Recent studies suggest a compensatory relationship between CDK4/6 and CDK2 under conditions of acute CDK2 inhibition (Fig. 7A; ref. 30). We have demonstrated that in addition to high cyclin E1 levels that lead to constitutive activation of CDK2, CDK4/6 must also be inactivated or inhibited for CDK2 to be essential for the G₁-S-phase transition (Fig. 7B). Therefore, high *CCNE1*, intact Rb signaling, and repressed CDK4/6 via high levels of p16 are predictive of sensitivity to BLU-222/CDK2i monotherapy.

Our data supporting the requirement of Rb for BLU-222 response in *CCNE1*-high cells (Fig. 3A, B, and D-M) are perhaps not surprising given the canonical understanding of CDK2-cyclin E-mediated cell-

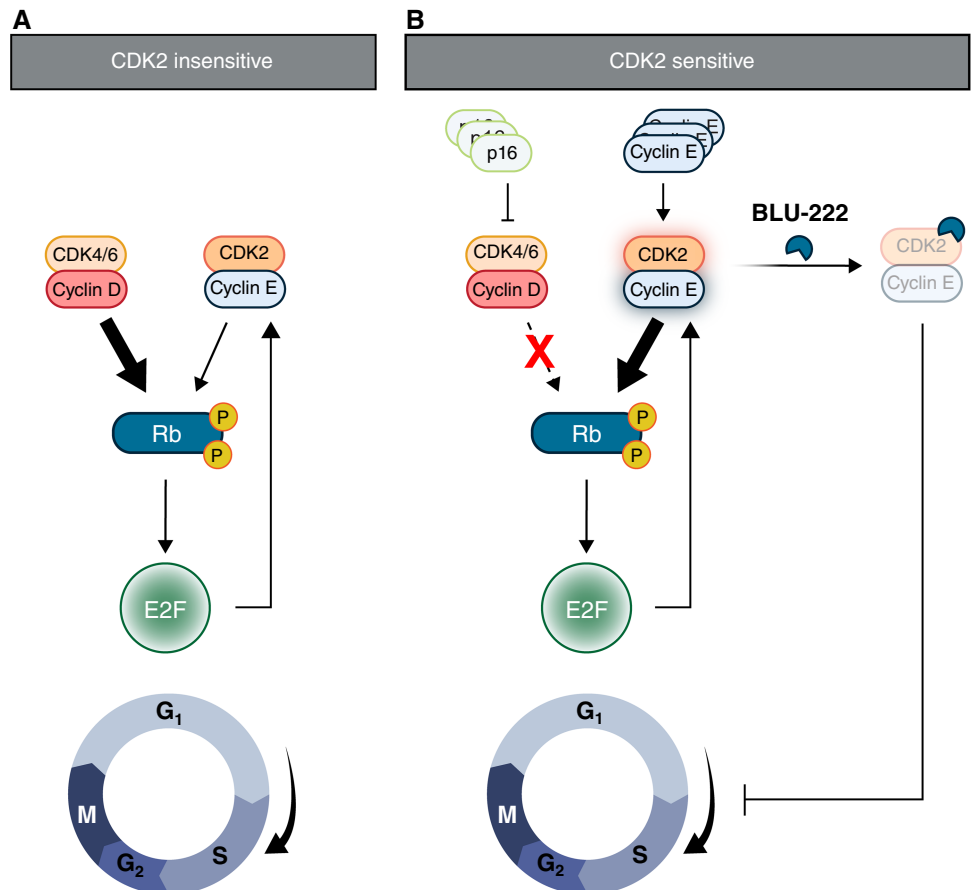
cycle progression through phosphorylation and inactivation of Rb activity. In the ovarian/endometrial setting, an intact Rb signaling circuit is required to mediate CDK2 dependency. In ovarian and endometrial cancers, loss of Rb is relatively rare (37). However, loss of Rb and its role as a potential resistance biomarker for CDK4/6is in hormone receptor (HR)-positive metastatic breast cancers has been previously reported (29, 38). This suggests that, although Rb is present, the Rb circuit can be inhibited to halt cell-cycle progression but that, upon loss of Rb, aberrant bypass mechanisms that are CDK2 independent are activated. Future work is needed to elucidate the mechanisms of CDK2- and Rb-independent cell-cycle progression in the cyclin E1-high cellular setting.

In addition to Rb, we identified the endogenous CDK4/6i p16 as critical in modulating BLU-222 sensitivity in *CCNE1*-high cells (Fig. 3A, C, and C-M). In a wild-type Rb setting, CDK2 compensates for low CDK4/6 activity when p16 is highly expressed, rendering cells vulnerable to pharmacologic CDK2 inhibition by BLU-222, whereas in cells with low p16 expression, CDK4/6 can promote the G₁-S-phase transition and is therefore a driver of resistance to CDK2 inhibitors. In BLU-222-treated ovarian cell lines, p16 (*CDKN2A*) transcript levels were not altered after short-term (24-hour) treatment (Supplementary Fig. S8), suggesting that p16 may not be a dynamic marker of resistance. Instead, basal p16 levels may dictate response to single-agent CDK2i.

Beyond single-agent activity for CDK2i, we demonstrated that BLU-222 can be used as a combination agent to induce strong antitumor activity. Importantly, we identified *CCNE1*/Rb⁺/low

Figure 7.

Model for BLU-222 response in a CDK2-dependent cell. **A**, In a CDK2-insensitive cell line, CDK4/6 is the major regulator of the G₁-S transition, with a nonessential role for CDK2. If CDK2 is inactivated in this setting, there is no effect on cell-cycle progression because CDK4/6 can compensate for loss of CDK2 activity and inactivate Rb. **B**, High levels of p16 render CDK4/6 inactive, and the G₁-S-phase transition becomes dependent on CDK2-mediated phosphorylation/inactivation of Rb. In this setting, cells are sensitive to pharmacologic inhibition of CDK2 by BLU-222 as a monotherapy to inhibit downstream activation of S-phase-related genes and arrest the cell cycle but are insensitive to pharmacologic inhibition of CDK4/6. Created in BioRender.com. House, N. (2025) <https://BioRender.com/u22y649>.



p16 levels as biomarkers predictive of response to BLU-222 in combination with CDK4/6i. Cotargeting CDK2 and CDK4/6 has been proposed to overcome aromatase inhibitor and CDK4/6 resistance in HR-positive breast cancers, in which *CCNE1* overexpression and aberrant CDK2 activation are hypothesized mechanisms of resistance (24, 39–43). In this setting, using a *CCNE1*-high, Rb-intact, p16-low combinatorial biomarker signature may further predict response to CDK2i with CDK4/6i.

We also demonstrated a role for BLU-222 as a chemosensitizer in *CCNE1*-high tumors using combination with standard-of-care chemotherapeutic agents. Analogous to CDK4/6i use in metastatic breast cancer in which CDK4/6is + endocrine therapy deepen efficacy (44), CDK2i with chemotherapeutic agents demonstrated a strong combination benefit in the *CCNE1*-high setting, converting static response to regression, or nonresponse to stasis with the combination in chemoresistant models (Fig. 6; Supplementary Fig. S15). We hypothesize that in the *CCNE1*-aberrant setting, CDK2-dependent replication fork stabilization combined with induced DNA damage sensitizes cells to the combination. Although CDK2i is inducing a G₁ arrest, which would prevent entry into the S phase, pRb inhibition is lessened between dosing intervals (Supplementary Fig. S3D). Therefore, in this scenario, cells that escape G₁ arrest and reenter the cell cycle would slip into the S phase between dosing intervals, leading to lethal fork collapse events and an antitumor response. The findings presented here warrant further study both to better understand the mechanism of combination activity and to assess whether the CDK2i chemosensitizing activity extends more broadly.

BLU-222 is currently being studied in the phase I/II clinical trial (VELA, NCT05252416) as a monotherapy in advanced solid tumors and in combination with ribociclib and fulvestrant in HR-positive/human EGFR2-negative breast cancer. Several selective CDK2 inhibitors, including PF-7104091 (45), INCB123667 (21), INX-315 (12), and BLU-222 (40), among others, have shown promising activity in preclinical studies and are in early clinical development. Using a combination of *in vitro*, *in vivo*, and bioinformatic approaches, here we have preclinically identified a novel combinatorial *CCNE1*/Rb/p16 biomarker signature to enrich for CDK2 dependency and sensitivity to CDK2i (e.g., BLU-222) in *CCNE1*-high ovarian and endometrial cancers. These biomarkers expand the mechanistic understanding of cell-cycle dysregulation that leads to aberrant CDK2 activity and may aid in the interpretation of emerging clinical data in *CCNE1*-high tumors. Furthermore, these data provide a

scientific rationale for potential future clinical trial patient selection criteria for monotherapy and combination therapy strategies, with potential beyond ovarian and breast cancers.

Authors' Disclosures

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Authors' Contributions

N.C. House: Conceptualization, data curation, supervision, validation, investigation, visualization, methodology, writing—original draft, writing—review and editing. **V.E. Brown:** Conceptualization, data curation, supervision, validation, investigation, visualization, methodology, writing—original draft, writing—review and editing. **M. Chen:** Data curation, software, formal analysis, validation, investigation, visualization, writing—original draft, writing—review and editing. **L. Yuan:** Validation, investigation, writing—review and editing. **S.L. Moore:** Validation, investigation, writing—review and editing. **J. Guo:** Formal analysis, writing—review and editing. **Y.J. Choi:** Conceptualization, supervision, validation, investigation, writing—review and editing. **L. Muthuswamy:** Formal analysis, supervision, writing—review and editing. **S. Ribich:** Conceptualization, writing—review and editing. **P. Ramsden:** Conceptualization, supervision, writing—review and editing. **K.L. Faia:** Conceptualization, supervision, writing—original draft, project administration, writing—review and editing.

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Note

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